

STP 470/Fall 2025
HW (pen-and-paper) *Naive Bayes, KNN, SVM*

1. **(30pts.)** Let $\mathbf{x}, \mathbf{z} \in \mathbb{R}^2$, $\mathbf{x} = (x_1, x_2)^T$ and $\mathbf{z} = (z_1, z_2)^T$. Consider the kernel

$$K(\mathbf{x}, \mathbf{z}) = (\mathbf{x}^T \mathbf{z} + 1)^2.$$

Show that

$$K(\mathbf{x}, \mathbf{z}) = (\Phi(\mathbf{x}))^T \Phi(\mathbf{z}),$$

where $\Phi : \mathbb{R}^2 \rightarrow \mathbb{R}^6$ is given by

$$\Phi(\mathbf{u}) = (1, \sqrt{2}u_1, \sqrt{2}u_2, u_1^2, u_2^2, \sqrt{2}u_1u_2).$$

2. **(30 pts.)** This problem is related to KNN and the *curse of dimensionality*.
- (a) Suppose we only have one feature (predictor) X , which in our notation translates to $p = 1$. Assume that X is uniformly distributed on $[0, 1]$. Associated with each observation is a response value. Suppose that we wish to predict a test observation's response using only observations that are within 10% of the range of X closest to that test observation. For instance, in order to predict the response for a test observation with $X = 0.6$, we will use observations in the range $[0.55, 0.65]$. On average, what fraction of the available observations will we use to make the prediction?
 - (b) Now suppose that we have a set of observations, each with measurements on $p = 2$ features X_1 and X_2 . We assume that (X_1, X_2) are uniformly distributed on $[0, 1] \times [0, 1]$. We wish to predict a test observation's response using only observations that are within 10% of the range of X_1 and within 10% of the range of X_2 closest to that test observation. For instance, in order to predict the response for a test observation with $X_1 = 0.6$ and $X_2 = 0.35$, we will use observations in the range $[0.55, 0.65]$ for X_1 and in the range $[0.3, 0.4]$ for X_2 . On average, what fraction of the available observations will we use to make the prediction?
 - (c) Now suppose that we have a set of observations on $p = 50$ features. Again the observations are uniformly distributed on each feature, and again the values of each feature are in $[0, 1]$. We wish to predict a test observation's response using observations within the 10% of each feature's range that is closest to that test observation. What fraction of the available observations will we use to make the prediction?
3. **(40 pts.)** This problem is related to the *Gaussian naive Bayes (GNB) classifier*. Let $\mathbf{X} = (X_1, \dots, X_p)^T$, such that each X_i is a continuous random variable. Assume that the response variable $Y \in \{0, 1\}$ follows a

Bernoulli distribution with parameter $q \in (0, 1)$. More precisely,

$$Pr(Y = 1) = q \text{ and } Pr(Y = 0) = 1 - q.$$

Further assume that for each $i = \overline{1, p}$, $(X_i|Y = 0)$ and $(X_i|Y = 1)$ follow Gaussian distributions, $(X_i|Y = 0) \sim N(\mu_{i0}, \sigma_i)$ and $(X_i|Y = 1) \sim N(\mu_{i1}, \sigma_i)$, $i = \overline{1, p}$. More precisely,

$$\begin{aligned} Pr(X_i = x_i|Y = 0) &= \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x_i - \mu_{i0})^2}{2\sigma_i^2}\right) \text{ and} \\ Pr(X_i = x_i|Y = 1) &= \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(x_i - \mu_{i1})^2}{2\sigma_i^2}\right), \quad i = \overline{1, p}. \end{aligned}$$

It is also assumed that within each class ($Y = 0$ and $Y = 1$) the predictors $X_1, \dots, X_p$ are independent.

(a) Compute the probability density functions

$$f_1(\mathbf{x}) = Pr(Y = 1|\mathbf{X} = \mathbf{x}) \text{ and } f_0(\mathbf{x}) = Pr(Y = 0|\mathbf{X} = \mathbf{x}),$$

where $\mathbf{X} = (X_1, \dots, X_p)^T$ and $\mathbf{x} = (x_1, \dots, x_p)^T$. Express your result, using the quantities defined below:

$$\begin{aligned} w_0 &= \ln \frac{1-q}{q} + \sum_{i=1}^p \frac{\mu_{i1}^2 - \mu_{i0}^2}{2\sigma_i^2} \\ w_i &= \frac{\mu_{i0} - \mu_{i1}}{\sigma_i^2}, \quad i = \overline{1, p}. \end{aligned}$$

(b) How does it compare to logistic regression? Justify your answer.

Hint. To ease your calculations, you may want to start as follows

$$\begin{aligned} Pr(Y = 1|\mathbf{X} = \mathbf{x}) &= \frac{Pr(Y = 1)Pr(\mathbf{X} = \mathbf{x}|Y = 1)}{Pr(Y = 1)Pr(\mathbf{X} = \mathbf{x}|Y = 1) + Pr(Y = 0)Pr(\mathbf{X} = \mathbf{x}|Y = 0)} \\ &= \frac{1}{1 + \frac{Pr(Y=0)Pr(\mathbf{X}=\mathbf{x}|Y=0)}{Pr(Y=1)Pr(\mathbf{X}=\mathbf{x}|Y=1)}} \\ &= \frac{1}{1 + \exp\left(\ln \frac{1-q}{q} + \ln \frac{Pr(\mathbf{X}=\mathbf{x}|Y=0)}{Pr(\mathbf{X}=\mathbf{x}|Y=1)}\right)} = \dots \end{aligned}$$